

OBSERVABLE COMPUTE FOUNDATION

South Dakota AI & Workforce Impact Report

State Employment Projections, AI Exposure, and Readiness Priorities

OCF Working Report | Version 1.0 | May 2026

Published by the Observable Compute Foundation

Rapid City, South Dakota | observablecompute.org

What This Report Is. What It Is Not.

This report is: • A South Dakota-focused synthesis of state labor projections and national AI impact evidence. • A readiness brief for funders, workforce boards, employers, and rural intermediaries. • A correction layer separating South Dakota DLR projections from national BLS AI case studies. • A practical map of where training, AI literacy, and internal mobility matter most.	This report is not: • A claim that AI has already caused mass unemployment in South Dakota. • A complete econometric model of all future job creation and displacement. • A claim that every occupation exposed to AI will lose employment. • A substitute for employer-specific workforce audits or worker-level transition planning.
--	---

Abstract

South Dakota enters the AI workforce transition from a position of projected labor market growth, not immediate statewide employment collapse. State projections show total employment rising from 511,117 jobs in 2022 to 550,566 jobs in 2032, a gain of 39,449 jobs and 7.72 percent overall growth. The state also projects 63,435 average annual openings, with replacement needs from exits and transfers accounting for nearly all demand. The evidence does not support a simple story of AI-driven mass job loss in South Dakota. It does support a sharper story: AI exposure is uneven, concentrated in office, financial, clerical, customer service, data, and professional workflows, while healthcare, technical services, construction, skilled trades, and data-intensive roles continue to grow. The policy issue is therefore readiness. South Dakota needs workers who can use, supervise, challenge, and adapt AI systems before the transition is forced on them by employers, vendors, or market pressure. The near-term risk is not that work disappears all at once. The risk is that entry-level pathways, routine office work, and low-digital-skill roles are quietly hollowed out while the training infrastructure arrives late.^{1,3,5}

Keywords: South Dakota workforce, AI exposure, workforce readiness, occupational projections, digital skills, rural access, Readiness Stack, ocf_schema_v1

At a Glance

South Dakota is growing. The exposure problem is not evenly distributed.

7.72% projected SD employment growth, 2022-2032 SD DLR	39,449 net jobs added by 2032 SD DLR	63,435 average annual openings SD DLR	93.8% openings from exits and transfers OCF calculation from DLR
62 occupations projected to decline SD DLR	+40.52% data scientist growth in SD SD DLR	+37.80% information security analyst growth in SD SD DLR	+15.2% professional and technical services sector growth SD DLR

Pipeline note: Statewide openings are dominated by replacement demand, not new job creation. That makes retraining and internal mobility as important as new-program creation.

I. Executive Summary

The cleanest conclusion is this: AI is not yet showing up in South Dakota as mass unemployment. It is showing up as task pressure, skill pressure, and uneven exposure. Statewide employment is projected to grow faster than the national baseline through 2032, but that does not mean every worker is safe. It means the first-order policy problem is transition capacity.^{2,8}

South Dakota is projected to add 39,449 jobs between 2022 and 2032. The state projects 63,435 annual openings, but only 3,945 of those openings are tied to annual employment growth. The rest come from exits and transfers. This matters because workforce systems cannot solve the AI transition by chasing new jobs alone. They have to help existing workers move inside changing occupations.^{1,3}

BLS AI case studies show the national pattern clearly: AI can reduce demand in some insurance, credit, legal-support, and routine analysis tasks while raising demand for software, database, cybersecurity, and AI-adjacent technical labor. South Dakota mirrors part of that pattern, but not perfectly. Some state-level projections remain positive in occupations that national AI analyses describe as exposed.⁵

II. Baseline: South Dakota Is Projected to Grow

DLR projects total South Dakota employment rising from 511,117 in 2022 to 550,566 in 2032. That is a numeric increase of 39,449 jobs and 7.72 percent growth. The September 2024 DLR labor bulletin rounds this to 7.7 percent and compares it with 2.8 percent national occupational growth over the same decade.^{1,2}

DLR also reports projections for 577 detailed occupations. Of those, 515 are projected to grow or remain stable, while 62 are projected to decline. That is the correct frame: South Dakota has declining roles, but the state-level data does not support a blanket claim that AI is producing broad labor-market contraction.²

The opening-demand structure is the little gremlin hiding under the spreadsheet. DLR projects 26,934 annual openings from labor force exits, 32,556 from occupational transfers, and 3,945 from annual employment growth. Replacement demand accounts for approximately 93.8 percent of total annual openings.³

III. What AI Is Actually Doing in the Data

AI exposure is not the same as AI job loss. Exposure means a job contains tasks reachable by AI. Vulnerability means exposure is likely to become displacement, wage pressure, or headcount reduction. Many public claims blur those categories, which produces bad policy and worse grant applications.^{8,9}

BLS now incorporates AI impacts through occupational case studies rather than a single all-purpose AI adjustment factor. That is the right approach for South Dakota because the same technology can reduce demand in one workflow while increasing demand in another. Software developers are exposed to AI, but BLS projects national growth because AI also increases demand for software solutions, data infrastructure, and AI system maintenance.⁵

Yale Budget Lab, updated April 2026, finds no economy-wide evidence that exposure, automation, or augmentation measures are yet related to broad employment or unemployment changes. That finding supports the South Dakota interpretation: monitor risk, build readiness, and avoid claiming a statewide job-loss wave before the data shows one.⁸

IV. Occupational Pressure Points

The table below separates South Dakota projections from national AI-impact projections. This correction matters. A national BLS decline for an occupation should not be presented as a South Dakota decline unless the state projection says the same thing.

Occupation	South Dakota DLR projection	National AI or automation context	OCF interpretation
Tellers	-9.20% from 2022 to 2032	BLS OOH projects -13% nationally from 2024 to 2034, citing online banking and automation.	Clear SD decline. Priority for digital banking, fraud, service, and advisory retraining.
Cashiers	+3.53% from 2022 to 2032, with 2,438 annual openings	BLS OOH projects -10% nationally from 2024 to 2034 due to self-checkout and online sales.	Not an immediate SD collapse. Still high exposure because task mix is changing.
Credit Analysts	+0.28% from 2022 to 2032	BLS AI case study projects -3.9% nationally from 2023 to 2033.	Small SD growth does not remove workflow pressure. Credit scoring and report synthesis are exposed.
Financial and Investment Analysts	+12.36% from 2022 to 2032	BLS projects +9.5% nationally from 2023 to 2033.	Likely augmentation, not replacement. Human judgment and fiduciary context remain valuable.
Insurance Appraisers, Auto Damage	+1.92% from 2022 to 2032	BLS projects -9.2% nationally	State projection differs from national AI case study.

		from 2023 to 2033.	Monitor claims-tech adoption.
Customer Service Representatives	-0.59% from 2022 to 2032	AI tools can automate routine inquiry handling and summarization.	Small decline, high workflow exposure. Best transition target for AI-assisted service training.
Software Developers	+26.82% from 2022 to 2032	BLS projects +17.9% nationally from 2023 to 2033.	High exposure and high growth. AI raises productivity and demand for implementation.
Data Scientists	+40.52% from 2022 to 2032	AI adoption increases demand for data quality, analysis, governance, and model evaluation.	Small base, high strategic value. Pipeline and retention issue.
Information Security Analysts	+37.80% from 2022 to 2032	Cybersecurity is a core AI-adjacent infrastructure need.	High-growth resilience role. Critical for public sector, health, finance, and small business.

Table 1. South Dakota occupational projections compared with national AI and automation context. Sources: SD DLR statewide occupational projections, BLS Monthly Labor Review, and BLS Occupational Outlook Handbook.

V. Growth and Augmentation Opportunities

The growth side of the ledger is not cosmetic. South Dakota projects strong growth in data scientists, information security analysts, software developers, financial and investment analysts, medical and health services managers, nurse practitioners, and industrial machinery mechanics. Those roles are not all safe from AI. Many are valuable because AI makes their work more central.^{1,2}

Industry projections reinforce the same pattern. Professional and Technical Services is projected to grow 15.2 percent. Health Care and Social Assistance is projected to grow 12.0 percent. Construction is projected to grow 11.5 percent. These sectors are not immune to AI, but they are less likely to experience simple substitution across the entire worker base.⁴

The South Dakota Science and Technology Plan 2030 identifies the need to grow a prepared STEM workforce and notes that STEM jobs in South Dakota are projected to grow 1.3 percent annually by 2032, compared with 0.7 percent for non-STEM jobs. That aligns with the DLR pattern: the higher-value transition is toward digital, technical, cybersecurity, data, and applied AI capability.¹⁰

VI. Geography: Rural Stability Can Hide Readiness Risk

Tufts Digital Planet projects that the Dakotas are among the least affected states in absolute AI job and income losses, while information, finance and insurance, and professional, scientific, and technical services show the steepest industry vulnerability nationally. That means South Dakota may face lower absolute exposure than large coastal states, but concentrated risk in office-heavy and finance-heavy local labor markets.⁹

For South Dakota, that points to a monitoring strategy rather than a panic strategy. Sioux Falls should be treated as the primary finance, insurance, and back-office workflow exposure zone.

Rapid City should be treated as a mixed exposure zone: healthcare, tourism, government, professional services, and small-business administration. Rural counties may appear safer by occupational mix, but they often have fewer training providers, weaker broadband access, and less employer-funded upskilling.^{9,10}

This is the rural trap: lower measured AI job risk can coexist with lower adaptation capacity. A worker in a large metro may face greater exposure, but also more training options and employer mobility. A rural worker may face slower exposure but have fewer ladders when the technology finally arrives.

VII. Training and Policy Signals

South Dakota DLR has already begun moving from abstract AI discussion to employer-facing implementation support. Its Workforce Knowledge Series session on AI tools for employers focused on practical AI uses, workforce effects, disability considerations, risk, and responsible adoption.¹¹

The federal workforce system is also shifting. U.S. DOL TEGL 03-25 encourages state and local workforce boards to use WIOA funding to help youth and adults develop AI skills. U.S. DOL TEN 07-25 provides an AI literacy framework for public workforce and education systems.^{12,13}

The skills data should be stated carefully. The claim that South Dakota has a 60 percent projected increase in technological skill demand should not be used unless a South Dakota-specific source is produced. A stronger and cleaner statement is that national research shows substantial growth in technological skill demand, with McKinsey estimating a 29 percent increase in U.S. technological skill demand by 2030 under its midpoint scenario.¹⁴

VIII. Data Confidence and Limits

Claim	Confidence	Reason
South Dakota employment grows 7.72 percent from 2022 to 2032.	0.95	Direct DLR statewide projection.
Average annual openings are 63,435, mostly from exits and transfers.	0.95	Direct DLR statewide projection and demand table.
AI is not yet causing broad statewide unemployment in South Dakota.	0.80	Supported by state growth projections and Yale national monitoring, but no SD-specific AI displacement dataset exists.
Office, finance, clerical, and customer-service workflows face higher AI exposure.	0.85	Supported by BLS AI case studies, BLS OOH automation discussion, and Tufts national industry vulnerability analysis.
Sioux Falls and Rapid City require closer monitoring than rural counties.	0.70	Reasoned inference from sector mix and regional labor-market structure, not a direct DLR AI exposure ranking.
Technological skills demand is rising sharply.	0.80	National research plus state STEM strategy. Not verified as SD-specific 60 percent.

Table 2. OCF confidence ratings. 0.90 and above means strong direct source support. 0.75 to 0.89 means moderate to strong support. Below 0.75 requires caution or further sourcing.

Implications

Funders should treat South Dakota as a readiness opportunity, not a post-displacement rescue case. The strongest use of capital is not emergency unemployment mitigation. It is early training infrastructure, employer-connected AI literacy, and transition support for office, finance, service, and administrative workers before displacement appears in unemployment data.

State workforce boards should build AI literacy into WIOA, adult education, youth pathways, and incumbent-worker training. The federal guidance already permits this direction. The missing piece is implementation design: modular credentials, employer-linked practice, and worker supports that do not assume every participant has equal broadband, device access, time, or confidence.^{12,13}

Rural intermediaries should focus on access as the binding constraint. Rural South Dakota does not need a slick AI bootcamp in one city. It needs distributed training through libraries, schools, employers, tribal colleges, community anchors, and hybrid models that treat transportation, devices, broadband, childcare, and trust as part of the program architecture.

What the Research Agrees On

1. AI exposure is uneven.

The highest-risk workflows are not evenly spread across the economy. Finance, information, professional services, customer support, clerical work, and routine analysis show greater exposure than many physical, care, and skilled-trade roles.

2. Exposure does not equal displacement.

A job can be highly exposed and still grow if AI increases demand for the product, service, infrastructure, or professional judgment attached to that work.

3. Replacement demand dominates South Dakota openings.

The state does not only need new-job creation. It needs mobility, retraining, and retention systems for workers already in motion.

4. AI literacy is becoming a workforce baseline.

Federal DOL guidance, state employer education, and STEM workforce planning all point in the same direction: workers need practical AI fluency, not vague AI awareness.

5. Rural access determines who benefits.

Rural communities may have lower absolute exposure but weaker response capacity. That makes access infrastructure a workforce issue, not a side issue.

The Bottom Line

South Dakota is not staring at an AI unemployment cliff in the official data. It is staring at a readiness cliff.

The state is projected to grow. That growth gives South Dakota time, but time is not the same thing as safety. AI pressure will arrive first through tasks: reports, claims, inquiries, scheduling, coding, credit review, document review, data entry, customer service, and routine office workflows.

The right response is not to promise workers that AI will leave them alone. It will not. The right response is to make sure South Dakota workers are not forced to meet AI through layoffs, vendor mandates, or silent task compression.

OCF should frame this as a workforce readiness mission: help South Dakota workers use AI before AI is used against their bargaining power, career mobility, and access to opportunity.

Sources Cited

Citation keys are abbreviated for readability. Full references include source organization, title, year, and URL where available.

Citation key	Full reference
1. SD DLR, 2024a	South Dakota Department of Labor and Regulation, Labor Market Information Center. South Dakota Occupational Employment Projections 2022-2032. June 21, 2024. https://dlr.sd.gov/lmic/documents/projections/occupational_projections_2022_2032_statewide_south_dakota.pdf
2. SD DLR, Sept. 2024	South Dakota Department of Labor and Regulation, Labor Market Information Center. South Dakota Occupational Employment Projections to 2032. South Dakota e-Labor Bulletin, September 2024. https://dlr.sd.gov/lmic/lb/2024/lbart_sept2024_occ_projections_fastest_growing.aspx
3. SD DLR, Oct. 2024	South Dakota Department of Labor and Regulation, Labor Market Information Center. Diving Deeper into Occupational Demand Projected to 2032. South Dakota e-Labor Bulletin, October 2024. https://dlr.sd.gov/lmic/lb/2024/lbart_oct2024_occ_demand_projections.aspx
4. SD DLR, July 2024	South Dakota Department of Labor and Regulation, Labor Market Information Center. South Dakota Industry Trends to 2032. South Dakota e-Labor Bulletin, July 2024. https://dlr.sd.gov/lmic/lb/2024/lbart_july2024_industry_projections_2022_2032.aspx
5. BLS MLR, 2025	U.S. Bureau of Labor Statistics. Incorporating AI Impacts in BLS Employment Projections: Occupational Case Studies. Monthly Labor Review, 2025. https://www.bls.gov/opub/mlr/2025/article/incorporating-ai-impacts-in-bls-employment-projections.htm
6. BLS OOH Tellers, 2025	U.S. Bureau of Labor Statistics. Occupational Outlook Handbook: Tellers. Last modified August 28, 2025. https://www.bls.gov/ooh/office-and-administrative-support/tellers.htm
7. BLS OOH Cashiers, 2025	U.S. Bureau of Labor Statistics. Occupational Outlook Handbook: Cashiers. Last modified August 28, 2025. https://www.bls.gov/ooh/sales/cashiers.htm
8. Yale Budget Lab, 2026	The Budget Lab at Yale. Tracking the Impact of AI on the Labor Market. Updated April 16, 2026. https://budgetlab.yale.edu/research/tracking-impact-ai-labor-market
9. Tufts Digital Planet, 2026	Digital Planet, The Fletcher School at Tufts University. Will Wired Belts Become the New Rust Belts? AI and the Emerging Geography of American Job Risk. 2026. https://digitalplanet.tufts.edu/ai-and-the-emerging-geography-of-american-job-risk-page/
10. SD S&T Plan, 2025	South Dakota EPSCoR REACH Committee and RTI International. South Dakota Science and Technology Plan 2030. Adopted March 11, 2025. https://sdbor.edu/wp-content/uploads/2025/09/SD-ST-Plan-Report_RTI_FINAL_03_11_25-2.pdf

11. SD DLR, Dec. 2025	South Dakota Department of Labor and Regulation. Workforce Knowledge Series: AI Tools for Employers. Employer Connection, December 2025. https://dlr.sd.gov/workforce_services/businesses/publications/employer_connection/dec-2025.html
12. U.S. DOL TEGL 03-25	U.S. Department of Labor, Employment and Training Administration. Training and Employment Guidance Letter No. 03-25: Encouraging the Use of WIOA Funding to Help Youth and Adults Develop Artificial Intelligence Skills. August 26, 2025. https://www.dol.gov/sites/dolgov/files/ETA/advisories/TEGL/2025/TEGL%2003-25/TEGL%2003-25.pdf
13. U.S. DOL TEN 07-25	U.S. Department of Labor, Employment and Training Administration. Training and Employment Notice No. 07-25: The U.S. Department of Labor Artificial Intelligence Literacy Framework. February 13, 2026. https://www.dol.gov/sites/dolgov/files/ETA/advisories/TEN/2025/TEN%2007-25/TEN%2007-25%20%28complete%20document%29.pdf
14. McKinsey MGI, 2024	McKinsey Global Institute. A New Future of Work: The Race to Deploy AI and Raise Skills in Europe and Beyond. May 21, 2024. https://www.mckinsey.com/mgi/our-research/a-new-future-of-work-the-race-to-deploy-ai-and-raise-skills-in-europe-and-beyond

Model Reference Appendix

Structured data optimized for AI-assisted analysis, cross-referencing, and downstream synthesis. Machine-parseable. Not intended as primary narrative.

```
{
  "paper_metadata": {
    "title": "South Dakota AI & Workforce Impact Report",
    "publisher": "Observable Compute Foundation",
    "principal_contributors": [
      "Julia Renee Hunsberger",
      "Adam I. Stratmeyer, J.D.",
      "Alex / GPT-5.5 Thinking (OpenAI)"
    ],
    "ein": "41-4747049",
    "publication_date": "2026-05",
    "version": "1.0",
    "schema": "ocf_schema_v1",
    "website": "observablecompute.org",
    "location": "Rapid City, South Dakota",
    "scope": "south_dakota_statewide_with_national_ai_context"
  },
  "ocf_named_constructs": {
    "readiness_stack": "Three-tier model: Tier 1 Foundational Readiness, Tier 2 Digital Readiness, Tier 3 AI Readiness.",
    "rural_amplification_effect": "Lower access to training, broadband, devices, and employer-sponsored mobility can compound AI transition risk in rural areas.",
    "access_as_binding_constraint": "Institutional access, not worker motivation, is the limiting condition for workforce readiness.",
    "exposure_not_displacement": "AI reachability of tasks must not be treated as proof of job loss without labor-market evidence."
  },
  "key_findings": [
    {
      "id": "F001",
      "claim": "South Dakota employment is projected to grow from 511,117 in 2022 to 550,566 in 2032, adding 39,449 jobs.",
    }
  ]
}
```

```

    "source": "SD DLR, 2024a",
    "confidence": 0.95
  },
  {
    "id": "F002",
    "claim": "South Dakota projects 63,435 average annual openings, with approximately 93.8 percent
from exits and transfers.",
    "source": "SD DLR, 2024a; OCF calculation",
    "confidence": 0.95
  },
  {
    "id": "F003",
    "claim": "DLR identifies 62 declining occupations out of 577 detailed occupations.",
    "source": "SD DLR, Sept. 2024",
    "confidence": 0.95
  },
  {
    "id": "F004",
    "claim": "AI is not yet visible as broad economy-wide unemployment in current public monitoring
data.",
    "source": "Yale Budget Lab, 2026",
    "confidence": 0.8
  },
  {
    "id": "F005",
    "claim": "South Dakota teller employment is projected to decline 9.20 percent from 2022 to 2032.",
    "source": "SD DLR, 2024a",
    "confidence": 0.95
  },
  {
    "id": "F006",
    "claim": "South Dakota data scientists and information security analysts are projected to grow
40.52 percent and 37.80 percent respectively.",
    "source": "SD DLR, 2024a",
    "confidence": 0.95
  },
  {
    "id": "F007",
    "claim": "National BLS AI case studies project software developers to grow 17.9 percent nationally
from 2023 to 2033 despite AI exposure.",
    "source": "BLS MLR, 2025",
    "confidence": 0.9
  },
  {
    "id": "F008",
    "claim": "The previous 60 percent technological-skills claim should not be stated as South Dakota-
specific without a direct state source.",
    "source": "OCF audit using McKinsey MGI, 2024 and SD DLR sources",
    "confidence": 0.8
  }
],
"what_works": {
  "employer_connected_ai_literacy": "Train workers on actual workplace tasks and tools, not abstract AI
awareness.",
  "modular_credentials": "Build stackable credentials across foundational, digital, and AI readiness.",
  "rural_hybrid_delivery": "Use distributed anchors such as schools, libraries, tribal colleges,
employers, and job service offices.",
  "internal_mobility": "Move exposed workers into adjacent roles before displacement forces job
search.",
  "transparent_labor_market_monitoring": "Separate exposure, augmentation, automation, wage pressure,
and actual employment change."
},
"what_frameworks_agree_on": [
  "AI exposure is uneven across occupations and industries.",
  "Exposure is not the same as displacement.",

```

```

    "Digital and AI literacy are becoming baseline workforce readiness requirements.",
    "South Dakota has growth capacity but must solve transition infrastructure.",
    "Rural access is a workforce constraint, not an optional equity add-on."
  ],
  "causal_chains": {
    "sd_ai_readiness_chain": [
      "AI adoption reaches office and analysis workflows",
      "Routine tasks compress or automate",
      "Workers without AI fluency lose mobility",
      "Employers hire or retain AI-capable workers",
      "Outcome: readiness gap becomes wage and opportunity gap"
    ],
    "rural_transition_chain": [
      "Lower absolute exposure creates delayed urgency",
      "Training infrastructure remains sparse",
      "AI tools arrive through vendors and employers",
      "Workers face transition with fewer supports",
      "Outcome: lower exposure can still produce higher hardship"
    ],
    "augmentation_growth_chain": [
      "AI raises productivity and infrastructure needs",
      "Demand grows for software, cybersecurity, data, and implementation roles",
      "Workers with technical and AI readiness capture gains",
      "Outcome: augmentation benefits are unevenly distributed"
    ]
  }
}

```

Authorship and Contributions

This paper was published by the Observable Compute Foundation (EIN 41-4747049), a South Dakota nonprofit focused on workforce readiness and technology access. Principal contributors: Julia Renee Hunsberger (strategic framing and OCF mission alignment), Adam I. Stratmeyer, J.D. (research architecture and editorial direction), and Alex / GPT-5.5 Thinking (OpenAI) (research synthesis, source audit, and document production). OCF is transparent about AI involvement in its research and publishing work. The analysis, editorial decisions, and OCF programmatic framing are those of the Foundation.

2026 Observable Compute Foundation. Open licensing. Cite with attribution.